

Conformal Score Prediction in Figure Skating Competitions

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Introduction. Predicting figure-skating scores is a challenging modelling problem because the International Skating Union (ISU) scoring system combines objective, rule-based evaluation of technical elements with subjective human judgement of artistic and performance quality. This study examines whether the final score of a skater can be estimated using only technical-element data, combining reliability (validity) and information (precision) of the prediction intervals produced by the Conformal Prediction framework for regression. In general, the results indicate that machine learning models can effectively learn scoring patterns in figure skating.

Data. The source of the data is the file `data/ml_data_sp.csv` from the repository Rooobeam <https://github.com/rooobeam/figure-skating/tree/master>

Out of 14 features presented in the repository, we used only 7 technical elements (1) gender; (2) bv (best value of technical elements); (3) num_quads; (4) num_triples; (5) num_doubles; (6) num_jump_errors; (7) num_spin_errors; – and the judge assessment: (8) judge_total as the label.

Task. The task is to estimate the value of the label for a new skater with his/her feature vector by training on older feature vectors and their labels.

Methodology. We use the kernelised Ridge Regression Confidence Machine [Nouretdinov et al. \(2001\)](#) and its Nearest Neighbour analogue developed later and presented in [Vovk et al. \(2005\)](#). According to the validity properties, the output of the algorithm is a prediction region (interval) that covers the true value with a pre-selected confidence level (90% or 99%).

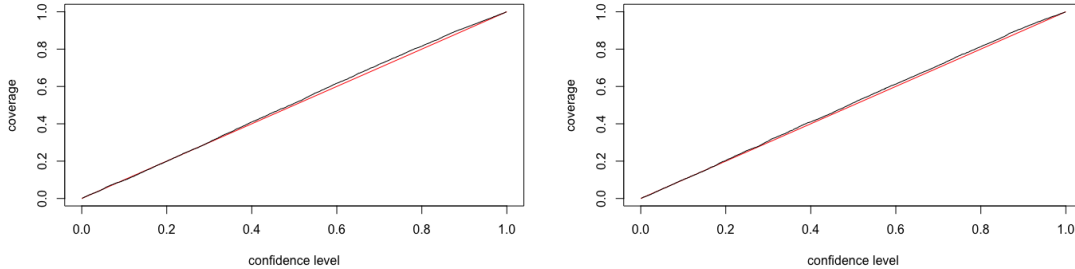
Evaluation. Success is reflected in the size (width) of the prediction set. The smaller the better, as this makes the prediction more informative (assuming that it is valid). A wider interval means a more uncertain (imprecise) prediction. The methods are applied in an online mode (training on the past), so the results are automatically averaged across different sample sizes.

Examples of individual predictions (gender = F). Each example is trained on the previous ones. E.g., the training sample size for No.3001 is 3000.

No.	bv	quads	triples	doubles	jump	spin	judge total	99%	90%
1001	28.83	0	2	2	1	0	51.95	[38;68]	[44;62]
2001	31.51	0	6	2	0	0	65.05	[55;79]	[59;75]
3001	20.80	0	2	2	1	0	43.81	[31;53]	[35;49]
4001*	22.48	0	2	2	1	0	43.11	[41;61]	[46;57]*
5001	29.03	0	2	2	3	0	49.56	[42;58]	[46;55]
6001	32.95	0	3	1	0	0	78.93	[71;87]	[74;83]
7001	35.99	0	3	0	2	0	69.18	[63;79]	[67;76]
8001	26.79	0	3	1	1	0	44.52	[41;57]	[45;54]
8801	19.95	0	1	2	1	1	50.06	[40;57]	[44;53]

*In the red case 4001, the true value of the label is covered by the 99% prediction set, but not by 90% prediction set.

Validity plots demonstrate empirical coverage for kNN (k=3), F (left), M (right):



Evaluation results show the advantage of k-Nearest-Neighbours over Ridge Regression for this application (the best parameters are shown).

Gender = F	99% width	Gender = M	99% width
RR (dot product)	25.30	RR (dot product)	34.74
RR (poly, deg.2)	25.10	RR (poly, deg.2)	34.65
kNN (k=3)	20.91	kNN (k=3)	27.52

Future research. In the next stage of this project, the data will include some subjective characteristics such as the nationalities of the skaters and judges.

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References

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